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mysql> -- Task 1 :-
Query OK, 0 rows affected (0.00 sec)

mysql> DROP DATABASE IF EXISTS databaseLab;
Query OK, 7 rows affected (0.06 sec)

mysql> CREATE DATABASE databaseLab;
Query OK, 1 row affected (0.00 sec)

mysql> DROP USER IF EXISTS '2401CS20'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE USER '2401CS20'@'localhost' IDENTIFIED BY '12345678';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT ALL PRIVILEGES ON databaseLab.* TO '2401CS20'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> USE databaseLab;
Database changed
mysql>
mysql> CREATE TABLE branch (
  ->   branch_name VARCHAR(20) PRIMARY KEY,
  ->   branch_city VARCHAR(20),
  ->   assets DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE customer (
  ->   customer_name VARCHAR(20) PRIMARY KEY,
  ->   customer_street VARCHAR(20),
  ->   customer_city VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE TABLE account (
  ->   account_no INT(10) PRIMARY KEY,
  ->   branch_name VARCHAR(20),
  ->   balance DECIMAL(10,2),
  ->   FOREIGN KEY (branch_name) REFERENCES branch(branch_name)
  -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE depositor (
  ->   customer_name VARCHAR(20),
  ->   account_no INT(10),
  ->   PRIMARY KEY (customer_name, account_no),
  ->   FOREIGN KEY (customer_name) REFERENCES customer(customer_name),
  ->   FOREIGN KEY (account_no) REFERENCES account(account_no)
  -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)
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mysql>
mysql> CREATE TABLE loan (
  ->   loan_no INT(10) PRIMARY KEY,
  ->   branch_name VARCHAR(20),
  ->   amount DECIMAL(10,2),
  ->   FOREIGN KEY (branch_name) REFERENCES branch(branch_name)
  -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> CREATE TABLE borrower (
  ->   customer_name VARCHAR(20),
  ->   loan_no INT(10),
  ->   PRIMARY KEY (customer_name, loan_no),
  ->   FOREIGN KEY (customer_name) REFERENCES customer(customer_name),
  ->   FOREIGN KEY (loan_no) REFERENCES loan(loan_no)
  -> );
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> INSERT INTO branch VALUES
  -> ('Downtown','Denver',920000.00),
  -> ('Denver','Denver',700000.00),
  -> ('North Town','Aurora',450000.00),
  -> ('Mesa','Mesa',310000.00),
  -> ('Central Plaza','Boulder',850000.00),
  -> ('West Central','Denver',660000.00);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO customer VALUES
  -> ('Sidhant','7th St','Denver'),
  -> ('Aman','Lake Rd','Aurora'),
  -> ('Ayush','Hill St','Mesa'),
  -> ('Rahul','Park Ave','Denver'),
  -> ('Raju','Elm St','Boulder'),
  -> ('Manish','Sunset Blvd','Aurora');
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO account VALUES
  -> (301,'Denver',6000.00),
  -> (302,'Denver',8200.00),
  -> (303,'Downtown',12500.00),
  -> (304,'Central Plaza',7000.00),
  -> (305,'Denver',9800.00),
  -> (306,'West Central',15500.00);
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO depositor VALUES
  -> ('Sidhant',301),
  -> ('Sidhant',303),
  -> ('Aman',302),
  -> ('Rahul',305),
  -> ('Raju',304),
  -> ('Manish',306);
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

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mysql>
mysql> INSERT INTO loan VALUES
  -> (201,'Downtown',3500.00),
  -> (202,'Downtown',2600.00),
  -> (203,'North Town',62000.00),
  -> (204,'Central Plaza',90000.00),
  -> (205,'Mesa',2500.00),
  -> (206,'West Central',78000.00),
  -> (207,'North Town',54000.00);
Query OK, 7 rows affected (0.00 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO borrower VALUES
  -> ('Sidhant',201),
  -> ('Aman',202),
  -> ('Ayush',205),
  -> ('Rahul',203),
  -> ('Raju',204),
  -> ('Manish',206),
  -> ('Aman',207);
Query OK, 7 rows affected (0.00 sec)
Records: 7  Duplicates: 0  Warnings: 0
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mysql>
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mysql> -- Task 2:-
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Query OK, 0 rows affected (0.00 sec)
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mysql>
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mysql> -- Q1: Find the loan number for each loan with an amount > 2000
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Query OK, 0 rows affected (0.00 sec)
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```
mysql> SELECT loan_no
-> FROM loan
-> WHERE amount > 2000;
```

loan_no
201
202
203
204
205
206
207

```
7 rows in set (0.00 sec)
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mysql>
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mysql> -- Q2: Find all borrowers who have taken loans with numbers 101, 102, or 105
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Query OK, 0 rows affected (0.00 sec)
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```
mysql> SELECT customer_name
-> FROM borrower
-> WHERE loan_no IN (201,202,205);
```

customer_name
Sidhant
Aman
Ayush

```
3 rows in set (0.00 sec)
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mysql>
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mysql> -- Q3: Find names of all customers who have a loan at "Downtown" branch and the loan amount
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Query OK, 0 rows affected (0.00 sec)
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```
mysql> SELECT borrower.customer_name, loan.amount
-> FROM borrower
-> JOIN loan ON borrower.loan_no = loan.loan_no
-> WHERE loan.branch_name = 'Downtown';
```

customer_name	amount
Sidhant	3500.00
Aman	2600.00

```
2 rows in set (0.00 sec)
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mysql>
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mysql>
mysql> -- Q4: Find names of all customers who have an account at "Denver" branch
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT DISTINCT depositor.customer_name
-> FROM depositor
-> JOIN account ON depositor.account_no = account.account_no
-> WHERE account.branch_name = 'Denver';
+-----+
| customer_name |
+-----+
| Sidhant      |
| Aman         |
| Rahul        |
+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- Q5: Find all loans from branches whose names contain 'Central' and amount between 50,000 and 100,000
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT loan_no, branch_name, amount
-> FROM loan
-> WHERE branch_name LIKE '%Central%'
-> AND amount BETWEEN 50000 AND 100000;
+-----+-----+-----+
| loan_no | branch_name | amount |
+-----+-----+-----+
| 204     | Central Plaza | 90000.00 |
| 206     | West Central | 78000.00 |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> -- Q6: Find names of customers who have more than one account (without COUNT)
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT DISTINCT d1.customer_name
-> FROM depositor d1
-> JOIN depositor d2
-> ON d1.customer_name = d2.customer_name
-> AND d1.account_no <> d2.account_no;
+-----+
| customer_name |
+-----+
| Sidhant      |
+-----+
1 row in set (0.00 sec)
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mysql>
mysql> -- Task 3:-
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE TABLE audit_log (
  ->   loan_no INT(10),
  ->   customer_name VARCHAR(20),
  ->   audit_status VARCHAR(20),
  ->   FOREIGN KEY (loan_no) REFERENCES loan(loan_no),
  ->   FOREIGN KEY (customer_name) REFERENCES customer(customer_name)
  -> );
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
mysql> INSERT INTO audit_log VALUES
  -> (203,'Rahul','Pending'),
  -> (207,'Aman','Pending'),
  -> (203,'Manish','Pending'),
  -> (205,'Ayush','Pending'),
  -> (205,'Sidhant','Pending');
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> -- Q1: Delete all records from audit log where branch is "Mesa"
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE audit_log
  -> FROM audit_log
  -> JOIN loan ON audit_log.loan_no = loan.loan_no
  -> WHERE loan.branch_name = 'Mesa';
Query OK, 2 rows affected (0.00 sec)

mysql>
mysql> -- Q2: Find unique customer names from audit_log who live in same city as the branch of the loan
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT DISTINCT audit_log.customer_name
  -> FROM audit_log
  -> JOIN loan ON audit_log.loan_no = loan.loan_no
  -> JOIN branch ON loan.branch_name = branch.branch_name
  -> JOIN customer ON audit_log.customer_name = customer.customer_name
  -> WHERE customer.customer_city = branch.branch_city;
+-----+
| customer_name |
+-----+
| Manish        |
| Aman          |
+-----+
2 rows in set (0.00 sec)

[mysql> exit
Bye

```